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Backseat driver: Auxiliary learning for autonomous racing

by

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Abstract

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Notation

γ	Discount rate
λ	GAE discount rate
r_t	Reward received at time t
$G_t = \sum_{i=t}^{T} \gamma^{i-t} \cdot r_i$	Discounted return at time t
$\mathcal{S}$	Set of all states
S_t	Random variable for state at time t
s_t	State at time t
$\mathcal{O}$	Observation space
O_t	Random variable for agent's observation at time t
o_t	Agent's observation at time t
$\mathcal{A}$	Action space
A_t	Random variable for action taken at time t
a_t	Observed action taken at time t
π_θ	Policy given parameters θ
$\pi_\theta(a_t \mid o_t)$	Probability density for action a_t given observation o_t and policy parameters θ
$\Delta_{\mathcal{Y}}(X)$	Set of all probability distributions over variable X with support $\mathcal{Y}$
$V^\pi(s) = \mathbb{E}_\pi[R_t \mid S_t = s]$	Value function for policy π at state s
$Q^\pi(s, a) = \mathbb{E}_\pi[R_t \mid S_t = s, A_t = a]$	Action-value function for policy π at state s and action a
$\hat{X}$	Estimator for random variable X

$\hat{x}$	Sample estimate of random variable X
$A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$	Advantage function for policy π at state s and action a
$\mathrm{Var}(X)$	Variance of random variable X
$\rho_0(s)$	Initial state distribution
$X \sim Y$	Random variable X is distributed according to distribution Y
$X \dot{\sim} Y$	Random variable X is approximately distributed according to distribution Y
$\mathbb{R}$	The set of real numbers
$\mathbb{E}[X]$	Expected value of random variable X
$\mathbb{E}_{x\sim p}[f(x)]$	Expected value of function $f(x)$ when x is distributed according to p
$N(y;\mu;\sigma^2)$	Gaussian probability density function with mean μ and variance σ^2 with evaluation point y

Acronyms and abbreviations

LiDAR	Light Detection and Ranging
MDP	Markov Decision Process
MAE	Mean Absolute Error
MSE	Mean Squared Error
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
RMSE	Root Mean Squared Error

Chapter 1

Introduction

1.1. Background and motivation

Autonomous racing is a challenging problem in autonmous robotics: the high speed, tight control margins, and short time available to recover from bad decisions makes for a demanding testbed to develop and evaluate autonomous driving algorithms. Studying autonomous racing is a proxy for development of autonomous driving algorithms, which have the potential to reduce severe and fatal accidents (e.g. severe lane-departure accidents [1]), improve traffic flow [2], and reduce emissions [3].

Generally, there exist two broad approaches to autonomous driving: end-to-end learning, where a model is trained to map sensor inputs directly to control outputs, and modular pipelines, where the problem is broken down into subproblems such as perception, planning, and control. End-to-end learning has the advantage of being able to learn a policy directly from data, without requiring hand-engineered features or intermediate representations. However, it can be difficult to interpret and debug, and may require large amounts of data to learn effectively. Modular pipelines, on the other hand, can be more interpretable and easier to debug, but may require more domain knowledge and engineering effort to design and tune [4].

Auxiliary learning is a technique involving training a model to perform additional (auxiliary) tasks alongside the primary task, in the hope that these tasks will help the model learn representations that are useful for the primary task. In some cases, it can be used to improve the performance of reinforcement learning models by providing additional supervisory signals during training that improve the model's learned representations [5], while also aiding in interpretability of the learned representations [4].

1.2. Problem statement & objectives

An end-to-end RL agent that drives from LiDAR alone has to learn a representation of privledged information it can't directly observe in the real world (e.g. its own velocity and slip, the curvature ahead, and its opponent's location in the case of head-to-head (H2H) racing). The agent may struggle to learn these representations early in training, when it has not yet learned to drive well enough to explore the environment effectively, which can lead to slow convergence.

However, since the privledged information is available in simulation, we can use auxiliary learning to provide additional supervisory signals to the agent during training, which may help it learn better representations of the privledged information and improve its performance.

The objectives of this Skripsie are to (a) select an appropriate simulation environment for autonomous racing, (b) implement a baseline end-to-end RL agent that drives from LiDAR alone, (c) gain an understanding of what representations the baseline agent natively encodes, (d) implement auxiliary learning to provide additional supervisory signals to the agent during training, and (e) evaluate the performance of the auxiliary learning agent against the baseline agent.

Chapter 2

Related literature

2.1. Markov Decision Processes (MDPs)

A Markov Decision Process (MDP) is a mathematical framework for modelling problems in which an agent must make a series of decisions in an environment. At each time step t, the environment is in some state $s_t \in \mathcal{S}$, which is a *complete* description of the environment at time t. The agent, however, only partially observes the environment; its observation $o_t \in \mathcal{O}$, where $\mathcal{O}$ is the observation space, is a function of the state s_t. In this case, the environment is said to be *partially observable*, and the problem is formally a *partially observable* MDP (POMDP). The information available in the state s_t that is not available in the observation o_t is called *privileged information*.

Based on its observation o_t, the agent chooses an action $a_t \in \mathcal{A}$. How the agent chooses is described by its *policy*, π: a rule that tells the agent what to do in every situation it might encounter. One can think of the policy as the agent's strategy or rulebook: given the current observation, it returns the action the agent will take. For example, a policy for a self-driving car might say "if I observe a red light ahead, brake". The policy may also be stochastic, assigning a probability to each action rather than always picking the same one; we write $a_t \sim \pi(\cdot \mid o_t)$ to mean the action is sampled from the policy. Note that we consider the policy to map observations to actions, compared to the classical mapping of states to actions in [6] and [7], since the agent cannot act on information it does not have. The policy is not part of the MDP itself: the policy describes the agent's behaviour, while the MDP describes the environment in which the agent operates.

Once the agent takes an action a_t, the environment responds by transitioning to a new state $s_{t+1} \in \mathcal{S}$, emitting a reward to the agent, $r_t \in \mathbb{R}$, and a new observation $o_{t+1} \in \mathcal{O}$. The reward is a scalar value that indicates how good the outcome of the agent's action is. It is a function of the action taken, and both the current state and the

successor state: $r_t = R(s_t, a_t, s_{t+1})$. Note that the reward depends on the true state, not on the agent's observation of it.

The agent may experience a "life" within the same MDP many times over, and each time it does so is called an *episode*. An episode starts with the environment in some initial state $s_0 \sim \rho_0$, and ends when the agent reaches a terminal state, or after a fixed number of time steps. The agent's objective is to learn a policy that maximises the expected *discounted return*, $\mathbb{E}_\pi[G_0]$, where

$$G_t = \sum_{i=t}^{T} \gamma^{i-t} r_i$$

is the discounted return from time step t, T is the time step at which the episode terminates, and the expectation is taken over all trajectories that can arise from ρ_0, the environment's dynamics, and the policy π. The policy that achieves this maximum is called the *optimal policy*, π^*.

The discount factor $\gamma \in [0, 1]$ controls how much the agent values future rewards relative to immediate ones. The return can be thought of as a weighted sum of all future rewards, in which a reward received k steps from now is worth only γ^k times as much as the same reward received immediately. A γ close to 0 makes the agent short-sighted, while a γ close to 1 makes it value long-term outcomes almost as much as immediate ones.

Formally, a POMDP is defined by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{P}, \mathcal{Z}, R, \rho_0, \gamma)$, where

- $\boldsymbol{\mathcal{S}}$ is the state space, which is the set of all possible states the environment can be in.
- $\boldsymbol{\mathcal{A}}$ is the action space, which is the set of all possible actions the agent can take. An action $a_t \in \mathcal{A}$ is a decision made by the agent at time t that (may) affect the environment.
- $\boldsymbol{\mathcal{O}}$ is the observation space, which is the set of all possible observations the agent can receive.
- $\boldsymbol{\mathcal{P} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})}$ is the transition function, which defines the probability of transitioning to a successor state s_{t+1} given the current state s_t and action a_t, formally $s_{t+1} \sim \mathcal{P}(\cdot \mid s_t, a_t)$.
- $\boldsymbol{\mathcal{Z} : \mathcal{S} \to \Delta(\mathcal{O})}$ is the observation function, which defines the probability of the agent receiving observation o_t when the environment is in state s_t, formally $o_t \sim \mathcal{Z}(\cdot \mid s_t)$.

- $\boldsymbol{R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}}$ is the reward function, which defines the reward received by the agent given the action taken and the successor state, formally $r_t = R(s_t, a_t, s_{t+1})$.
- $\boldsymbol{\rho_0 \in \Delta(\mathcal{S})}$ is the initial state distribution, formally $s_0 \sim \rho_0$.
- $\boldsymbol{\gamma \in [0, 1]}$ is the discount factor described above.

Transitions satisfy the *Markov property*: the successor state depends only on the current state and action, and not on the rest of the history,

$$P(s_{t+1} \mid s_t, a_t, s_{t-1}, a_{t-1}, ..., s_0, a_0) = P(s_{t+1} \mid s_t, a_t).$$

This property holds for the states, but generally not for the observations: a single observation o_t does not, in general, contain enough information to predict o_{t+1}.

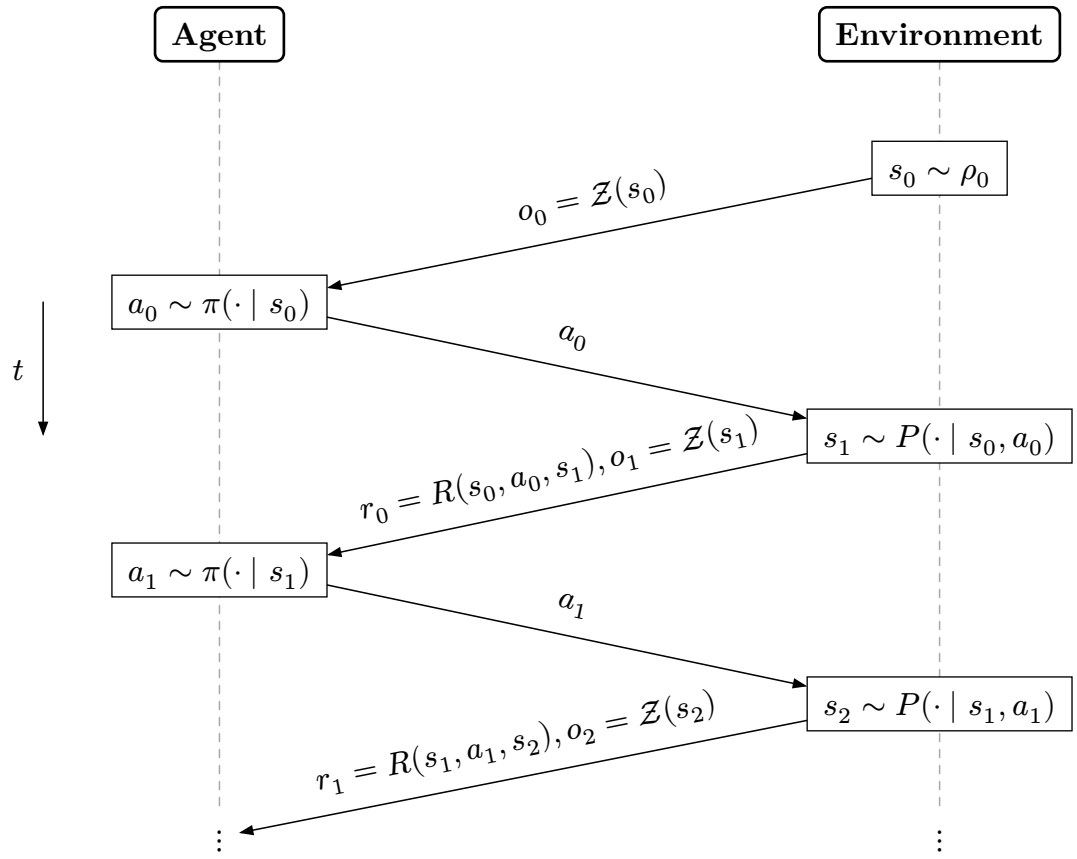


Figure 2.1: Interaction between the agent and the environment over time (t).

2.1.1. Value functions

Value functions are a central concept in reinforcement learning, providing a measure of how good it is for the agent to be in a particular state or to take a particular action. The state-value function,

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$

represents the expected discounted return G_t the agent can achieve starting from state s and following policy π thereafter. The action-value function,

$$Q^\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

represents the expected discounted return the agent can achieve starting from state s, taking action a, and following policy π thereafter. The advantage function,

$$A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s) \tag{1}$$

measures the relative value of taking action a in state s compared to the average value of all actions in that state.

These state-value and action-value functions satisfy the Bellman equations, which express the relationship between the value of a state and the values of its successor states. [6] The Bellman equation for the state-value function is given by:

$$V^\pi(s) = \mathbb{E}_\pi\left[R_{\{t+1\}} + \gamma V^\pi\left(S_{\{t+1\}}\right) \mid S_t = s\right] \tag{2}$$

and the Bellman equation for the action-value function is given by:

$$Q^\pi(s,a) = \mathbb{E}_\pi\left[R_{\{t+1\}} + \gamma Q^\pi\left(S_{\{t+1\}}, A_{\{t+1\}}\right) \mid S_t = s, A_t = a\right] \tag{3}$$

These Bellman equations Equation 2 and Equation 3 provide a recursive relationship that allows the agent to update its value estimates based on the observed rewards and the estimated values of successor states.

2.2. Reinforcement Learning (RL)

Reinforcement learning is a subfield of machine learning, where a policy π_θ is learned through an agent's interactions with an environment with the objective of maximising the expected discounted return under policy π_θ, $\mathbb{E}_{\pi_\theta}[G_t]$, by varying the parameters θ (i.e. finding the optimal policy π^*)[1]. This is done by sampling trajectories of the agent's interactions with the environment, and using the observed rewards to update the policy parameters θ in a way that increases the probability of actions that lead to higher returns.

RL is distinct from supervised learning in that during training, the optimal or correct action is not known - the agent must explore the environment and learn from the consequences of its actions. It is also distinct from classical unsupervised learning

[1]The parameterisation restricts the search to the policy class $\Pi_\Theta = \{\pi_\theta : \theta \in \Theta\}$, the set of policies the chosen model can represent. The optimal policy π^* need not lie in Π_Θ: for example, if π_θ outputs a Gaussian over actions, it cannot represent an optimal policy that is multimodal or deterministic. In that case, RL can at best find the policy in Π_Θ with the highest expected return, π_{θ^*} with $\theta^* = \arg\max_\theta \mathbb{E}_{\pi_\theta}[G_0]$.

methods in that there is a clear objective (maximising expected return) and feedback (rewards) to guide the learning process.

There exist two broad classes of RL algorithms: value-based methods and policy-based methods. Value-based methods learn a value function that estimates the expected return for each state or state-action pair ($\hat{V}(s_t)$ or $\hat{Q}(s_t, a_t)$ respectively), and derive a policy from this value function (for example, acting greedily in state s, choosing action $a = \arg\max_{a \in \mathcal{A}} \hat{Q}(s, a)$). Policy-based methods learn a parameterised policy directly, without explicitly estimating a value function. We focus on policy-based methods in this report, as value-based methods are less suitable for continuous action spaces [8] and policy-based methods can natively learn stochastic policies, without tricks such as ε-greedy. [9]

2.3. Policy gradient methods

Policy gradient methods are a class of RL training algorithms that learn a policy π_θ with parameters θ directly by maximising the expected return $\mathbb{E}[G_0]$ with respect to the policy parameters θ. Given a trajectory $\tau = (s_0, o_0, a_0, r_0, s_1, o_1, a_1, r_1, ..., s_{T-1}, o_{T-1}, a_{T-1}, r_{T-1}) \sim p_\theta$, where p_θ is the probability of the trajectory when following the policy π_θ, we intend to maximise the expected return with respect to the policy parameters θ:

$$J(\theta) = \mathbb{E}_{\tau \sim p_\theta}[G_0] = \mathbb{E}_{\tau \sim p_\theta}\left[\sum_{t=0}^{T} \gamma^t r_t\right] = \int p_\theta(\tau) G(\tau) \, \mathrm{d}\tau$$

Using the *log-derivative* trick, $\nabla_\theta p_\theta(\tau) = p_\theta(\tau) \nabla_\theta \log p_\theta(\tau)$, and assuming that the integral and derivative can be interchanged[2], the gradient of the expected return with respect to θ is then

$$\nabla_\theta J(\theta) = \nabla_\theta \int p_\theta(\tau) G(\tau) \, \mathrm{d}\tau = \int \nabla_\theta p_\theta(\tau) G(\tau) \, \mathrm{d}\tau = \int p_\theta(\tau) \nabla_\theta \log p_\theta(\tau) G(\tau) \, \mathrm{d}\tau$$

$$= \mathbb{E}_{\tau \sim p_\theta}[\nabla_\theta \log p_\theta(\tau) G(\tau)]$$

Considering that the probability of a trajectory can be factorised as

$$p_\theta(\tau) = p(s_0) \prod_{t=0}^{T-1} \underbrace{p(o_t \mid s_t)}_{\mathcal{Z}} \cdot \pi_\theta(a_t \mid o_t) \cdot \underbrace{p(r_t \mid s_t, a_t, s_{t+1}, o_t)}_{\mathcal{R}} \cdot \underbrace{p(s_{t+1} \mid s_t, a_t)}_{\mathcal{P}}$$

since $p(s_0)$, $\mathcal{P}$, $\mathcal{Z}$ and $\mathcal{R}$ do not depend on θ, their gradients vanish, so

[2] This can be done only under certain regularity conditions not explored in this report, but generally if the functions are continuous and differentiable *almost* everywhere, and the integral is finite.

$$\nabla_\theta \log p_\theta(\tau) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid o_t)$$

Then, the *policy gradient* can be expressed as

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid o_t) G(\tau) \right]$$

Write $G(\tau) = G_0$. Splitting the return at time t,

$$G_0 = \sum_{k=0}^{t-1} \gamma^k r_k + \gamma^t G_t, \quad G_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k,$$

where G_t is the discounted *reward-to-go*,

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t \mid o_t) \left(\sum_{k=0}^{t-1} \gamma^k r_k + \gamma^t G_t \right) \right]$$

However, since the expectation of the fixed past-reward terms is zero, they can be removed from the policy gradient without introducing bias. This can be seen by conditioning on the history $h_t = (s_0, o_0, a_0, r_0, ..., s_{t-1}, o_{t-1}, a_{t-1}, r_{t-1}, s_t, o_t)$:

$$\mathbb{E}_{a_t \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a_t \mid o_t) \mid h_t] = \sum_a \nabla_\theta \pi_\theta(a \mid o_t) = \nabla_\theta \sum_a \pi_\theta(a \mid o_t) = \nabla_\theta 1 = 0,$$

The past-reward terms therefore vanish in expectation, leaving

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} \left[\sum_{t=0}^{T-1} \gamma^t \nabla_\theta \log \pi_\theta(a_t \mid o_t) G_t \right].$$

Removing these zero-mean terms typically reduces the variance of the Monte Carlo estimate. In practice the factor γ^t is usually dropped; this yields a biased estimate of $\nabla_\theta J(\theta)$ for $\gamma < 1$ [10].

2.3.1. REINFORCE

The REINFORCE algorithm for policy learning was introduced by Williams in 1992 in the journal *Machine Learning* [11]. It is one of the simplest policy gradient methods: by sampling N trajectories $\tau^{(i)} \sim \pi_\theta$ of length T_i timesteps each, the policy gradient can be estimated in a Monte-Carlo fashion:

$$\widehat{\nabla_\theta J}(\theta) = \frac{1}{N \cdot \overline{T}} \sum_{i=1}^{N} \sum_{t=0}^{T_i - 1} \log\Big(\pi_\theta\Big(a_t^{(i)} \mid o_t^{(i)}\Big)\Big) \cdot G_t^{(i)}$$

where $\overline{T}$ is the average trajectory length. In order to run REINFORCE in an AutoGrad engine (such as PyTorch or TensorFlow), a loss function $\mathcal{L}_{\text{REINFORCE}}$ is defined such

that its gradient with respect to the policy parameters θ is equal to the policy gradient estimate above. This loss function is given by:

$$\mathcal{L}_{\text{REINFORCE}} = -\frac{1}{N \cdot \overline{T}} \sum_{i=1}^{N} \sum_{t=0}^{T_i - 1} \log\Big(\pi_\theta\Big(a_t^{(i)} \mid o_t^{(i)}\Big)\Big) \cdot G_t^{(i)}$$

Although REINFORCE is simple to implement, it has several drawbacks. First, it has high variance in its gradient estimates, which can lead to slow convergence and instability during training. Second, it does not make use of a value function to estimate the expected return, which can further increase variance. Finally, it is an on-policy algorithm, meaning that it can only learn from trajectories generated by the current policy, which can limit exploration and learning efficiency (since it cannot train multiple epochs on the same rollout).

2.3.2. Proximal Policy Optimization (PPO)

PPO is a modern algorithm for policy learning introduced in 2017 by researchers at OpenAI [12]. It improves on REINFORCE in two ways. First, it is an actor-critic method: alongside the policy, it learns a value function $V^\pi(s)$, which is used to compute an advantage estimate $\hat{A}_t$ and so reduces the variance of the policy gradient estimate. Second, it optimises a *clipped surrogate objective*, $\mathcal{L}_{\text{PPO}}$, that limits how far each update can move the policy. This prevents large, destructive steps and makes several epochs of updates on the same rollout possible.

Recall that the advantage function in Equation 1 is the difference between the action-value and state-value functions. It therefore measures how much better (positive) or worse (negative) an action is than the policy's average action in that state. In training, $\hat{A}_t(s_t, a_t)$ is estimated from observed rewards and the learned value function, $\hat{V}^\pi(s)$, for example with Generalized Advantage Estimation [13]. In PPO, the clipped surrogate objective is given by

$$\mathcal{L}_{\text{PPO}} = -\frac{1}{N \cdot \overline{T}} \sum_{i=1}^{N} \sum_{t=0}^{T_i - 1} \min\Big(r_t(\theta) \cdot \hat{A}_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon) \cdot \hat{A}_t\Big)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t \mid o_t)}{\pi_{\theta_{\text{old}}}(a_t \mid o_t)}$ is the ratio of the new policy's probability of taking action a_t given observation o_t to the old policy's, and ε is a hyperparameter that controls how much the policy is allowed to change in a single update. The clip function ensures that a policy update that would make $r_t(\theta)$ fall outside of the interval $[1-\varepsilon, 1+\varepsilon]$ is clipped to the nearest endpoint of that interval. This prevents large policy updates that could destabilise training. The min function on the other hand ensures that, while the policy's

update is capped when an update makes a good action ($A_t > 0$) more likely, it is not protected from penalties if an update makes a bad action ($A_t < 0$) more likely.

2.4. Auxiliary Learning in RL

In RL, the *primary task* is to update a policy π_θ to maximises the expected discounted return $\mathbb{E}_{\tau\sim\pi_\theta}[G_0]$. *Auxiliary learning* augments this with *auxiliary tasks* whose learning signals are used only to improve the agent's representations. Unline general multi-task learning, perfromance on the auxiliary tasks is not itself the goal, and their outputs are typically discarded at inference time. Because auxiliary tasks can supply training signal when the reward is sparse or delayed [14], they can improve sample efficiency, and, by acting as an inductive bias[3], generalisation [16]. We consider two formulations: auxiliary **losses**, which leave the RL objective unchanged and shape only the shared representation, and auxiliary **rewards**, which modify the objective itself and can therefore change the optimal policy.

2.4.1. Auxiliary Losses

In this formulation, the agent is trained on a weighted sum of the primary loss and K auxiliary losses. Each loss is computed by a separate *auxiliary head* that reads from a representation shared with the policy. Writing the parameters as $\theta = \left(\theta_{\mathrm{s}}, \theta_\pi, \theta_{\mathrm{a}}^{(1)}, ..., \theta_{\mathrm{a}}^{(K)}\right)$, where θ_{s} are the shared (encoder) parameters, θ_π the policy head and $\theta_{\mathrm{a}}^{(k)}$ the k-th auxiliary head, the total loss is

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{primary}}(\theta_{\mathrm{s}}, \theta_\pi) + \sum_{k=1}^{K} w^{(k)} \mathcal{L}_{\text{aux}}^{(k)}\left(\theta_{\mathrm{s}}, \theta_{\mathrm{a}}^{(k)}\right),$$

where $w^{(k)} \geq 0$ weights each auxiliary task relative to the primary loss (only relative weights matter, so the primary weight is fixed to 1). The gradients with respect to each parameter group are

$$\nabla_{\theta_\pi} \mathcal{L}_{\text{total}} = \nabla_{\theta_\pi} \mathcal{L}_{\text{primary}},$$

$$\nabla_{\theta_{\mathrm{a}}^{(k)}} \mathcal{L}_{\text{total}} = w^{(k)} \nabla_{\theta_{\mathrm{a}}^{(k)}} \mathcal{L}_{\text{aux}}^{(k)}, \quad \nabla_{\theta_{\mathrm{s}}} \mathcal{L}_{\text{total}} = \nabla_{\theta_{\mathrm{s}}} \mathcal{L}_{\text{primary}} + \sum_{k=1}^{K} w^{(k)} \nabla_{\theta_{\mathrm{s}}} \mathcal{L}_{\text{aux}}^{(k)}.$$

Only the shared parameters θ_{s} receive gradients from both the primary and auxiliary losses, which is how the auxiliary tasks influence the policy: they shape the representation it is built on

[3] Inductive bias is the set of assumptions a learning algorithm makes to predict outputs given inputs that it has not encountered before. [15]

The primary loss, such as $\mathcal{L}_{\text{PPO}}$, is still a policy-gradient surrogate with a high-variance gradient estimate. Auxiliary losses, by contrast, are usually ordinary supervised or self-supervised objectives (for example predicting the next observation, or the immediate reward), so their gradients are exact and low-variance, and do not depend on the reward being observed [citation]. This is what makes them useful when rewards are sparse or delayed. Their cost is that the auxiliary and primary gradients can conflict on the shared parameters, θ_{s} so the weights $w^{(k)}$ must be chosen with care.

2.4.2. Auxiliary Rewards/Shaped Rewards

Alternatively, the auxiliary task can be incorporated into the reward. The agent is given the modified reward

$$r_t^{\text{total}} = r_t^{\text{primary}} + \sum_{k=1}^{K} w^{(k)} r_t^{\text{aux},(k)},$$

where each $r_t^{\text{aux},(k)}$ rewards behaviour that is expected to help with the primary task, such as an intrinsic curiosity bonus [17] or progress towards a subgoal. The policy is then trained to maximise the expected return of r_t^{total} with the same policy gradient machinery as before.

Unlike auxiliary losses, this changes the objective being optimised: the agent now solves a different (PO)MDP, whose optimal policy need not be the same as π^* for the primary reward. Poorly chosen auxiliary rewards can therefore lead the agent to exploit them at the expense of the primary task (*reward hacking*). The exception is *potential-based shaping*, where the auxiliary reward has the form $\gamma\Phi(s_{t+1}) - \Phi(s_t)$ for some potential function Φ, which is guaranteed to leave the optimal policy unchanged [18].

The auxiliary rewards do influence the policy, but only indirectly through the return G_t, and therefore through the same high-variance score-function estimator as the primary reward. No exact gradient is available, and the agent must work out through its own exploration which auxiliary rewards actually contribute to the primary task. This credit-assignment problem is harder still when the auxiliary rewards are themselves sparse or delayed.

2.5. Loss scaling

When composing the L_{total}, the question naturally arises: what should be the values of the weights $\{w^{(1)}, ..., w^{(N)}\}$? Since the values cannot simply be learnt by gradient

descent (they would simply shrink to 0 or go negative to lower $\mathcal{L}_{\text{total}}$), we must use another technique.

2.5.1. Hyperparameter tuning

One may empirically estimate the optimal weights using hyperparameter tuning tools such as Optuna [19], or a simple grid search. However, when it is expensive to train a model, this method of search quickly becomes impractical, since every candidate set of weights requires its own training run.

Hyperparameter tuners differ in how they choose which weights to try next. Grid search evaluates every combination on a fixed N-dimensional grid, so its cost grows exponentially with the number of weights N. Random search samples configurations at random and is often more efficient than grid search, because usually only a few hyperparameters matter much [20]. Bayesian optimisation methods go further by building a model of how the score depends on the hyperparameters and using it to choose promising candidates. Examples include Gaussian-process-based optimisation [21] and the tree-structured Parzen estimator (TPE) [22], which is the default sampler in Optuna. Methods such as Hyperband [23] and its asynchronous variant ASHA [24] give each trial a small budget and only continue the best-performing ones. However, this method may kill RL trials that perform more exploration early on, so are not ideal for such scenarios.

Even with these savings, each trial still involves training a model, and the number of trials needed grows with the number of loss terms N. For multi-task models that are expensive to train, this motivates methods that learn the weights during training instead.

2.5.2. Uncertainty weighting

Two kinds of uncertainty can be modelled in a model's predictions [25]:

- *Epistemic* uncertainty: the model is unsure because it hasn't been trained on enough data. More data reduces epistemic uncertainty.
- *Aleatoric* uncertainty: the data itself is noisy or incomplete, so even a perfect model can't predict exactly. In this case, more training data doesn't help.

Aleatoric uncertainty has two forms:

- *Data-dependent* or *Heteroscedastic*: the noise changes from input to input (e.g. a blurry image is harder to make inference from than a clear one).

- *Task-dependent* or *Homoscedastic*: the noise is the same for every input and depends only on the task

Kendall et al. propose a solution to determining the weights $\{w_1, ..., w_N\}$ problem by modelling the homoscedastic (task-dependent) uncertainty only [26].

Instead of formulating the prediction for task i, $\hat{y_i} = f_i(\boldsymbol{x})$, as a point estimate, it is modelled as a Gaussian with mean $f_i(x)$ and variance σ_i^2:

$$\hat{Y}_i \sim p(y_i \mid f_i(x)) = N(y_i; f_i(x); \sigma_i^2)$$

The negative log-likelihood of this Gaussian probability density function evaluated at the true value y_i is then the loss value for task i:

$$L_i = -\log p(y_i \mid f_i(x)) = \frac{1}{2\sigma_i^2} \|y_i - f_i(x)\|_2^2 + \log \sigma_i$$

A similar derivation can be made for L1 loss functions by assuming the predictions are $Y_i \sim \text{Laplace}(f_i(x), b_i)$:

$$L_i = -\log p(y_i \mid f_i(x)) = \frac{1}{b_i} \|y_i - f_i(x)\|_1 + \log b_i$$

Then, assuming the tasks are independent, the total loss is the sum of the individual losses. The log-variance $\log \sigma_i$ or log-scale parameter $\log b_i$ is a learnable parameter that is updated during training, allowing the model to learn the optimal weights for each loss term.

Of course, in reality, the tasks are not independent, but this method has empirically been shown to work well in practice [26]. With some caveats like the fact that it produces a bias towards easy tasks, since they result in smaller σ_i values. This has led to the development of other methods not explored in this Skripsie, such as so-called Optimal Uncertainty Weighting [27].

2.6. Huber loss

The Huber loss (with parameter β or in some literature k) is a loss function that is less sensitive to outliers compared to the standard Mean Squared Error (MSE) L2 loss [28]. It is defined as:

$$\mathcal{L}^{\beta}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 \text{ if } |y - \hat{y}| \leq \beta \\ \beta \ |y - \hat{y}| - \frac{1}{2}\beta^2 \text{ otherwise} \end{cases}$$

In our case, since we have various loss terms, each relating to a different task where the scale of y differs, we must carefully choose the parameter β and scale each loss term appropriately.

Rewriting the loss to $\mathcal{L}^{\beta}\left(\frac{y-\mu}{\sigma}, \hat{y}\right)$ allows us to reduce the model's bias towards learning tasks with larger scales, since the loss is now computed on a normalised scale.

In actuality, PyTorch's implementation of the Huber loss (`smooth_l1_loss`) uses a rescaled definition, $\frac{\mathcal{L}^{\beta}(y,\hat{y})}{\beta}$, to prevent the β parameter from affecting the gradient in the linear region.

Chapter 3

System overview

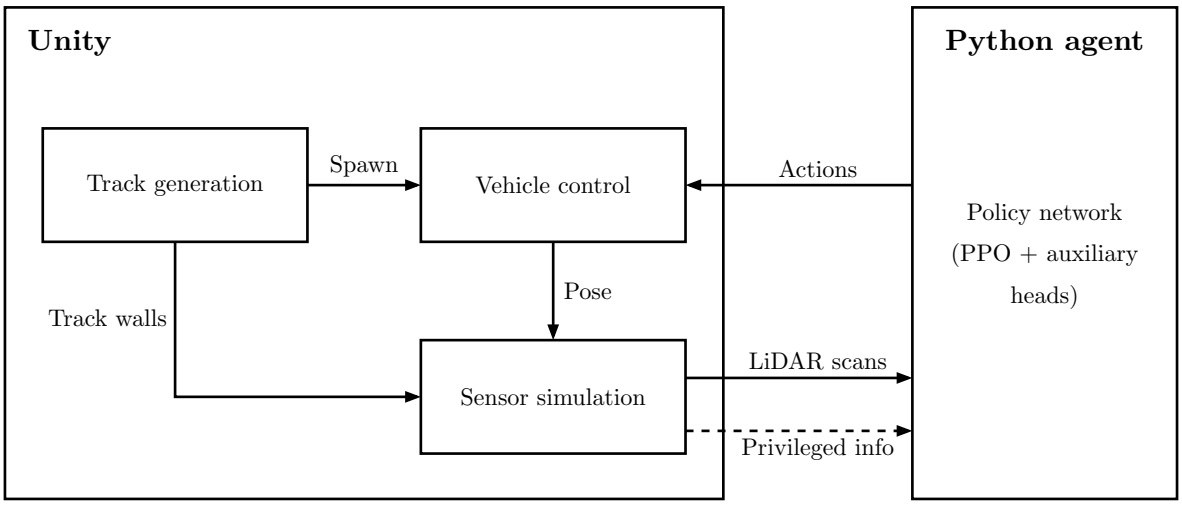


Figure 3.1: Subsystem diagram: the Unity simulation and the Python agent.

3.1. RoboRacer

RoboRacer (formerly F1TENTH) is an open-source 1/10th scale race car platform. It is an open standard used for learning and research in the field of autonomous racing, and with over 1000 related publications, it is widely used in the research community. [29] The RoboRacer race car (Figure 3.2) is a small vehicle governed by RoboRacer's rules [30], which includes specifications on the vehicle's dimensions, weight, and performance. During training, this vehicle is simulated with a wheelbase of 324 mm, a track width of 236 mm, and a maximum steering angle of 30° . The vehicle is powered by an electric motor and is equipped with a LiDAR sensor and camera for perception, as well as an onboard computer for processing and decision-making.

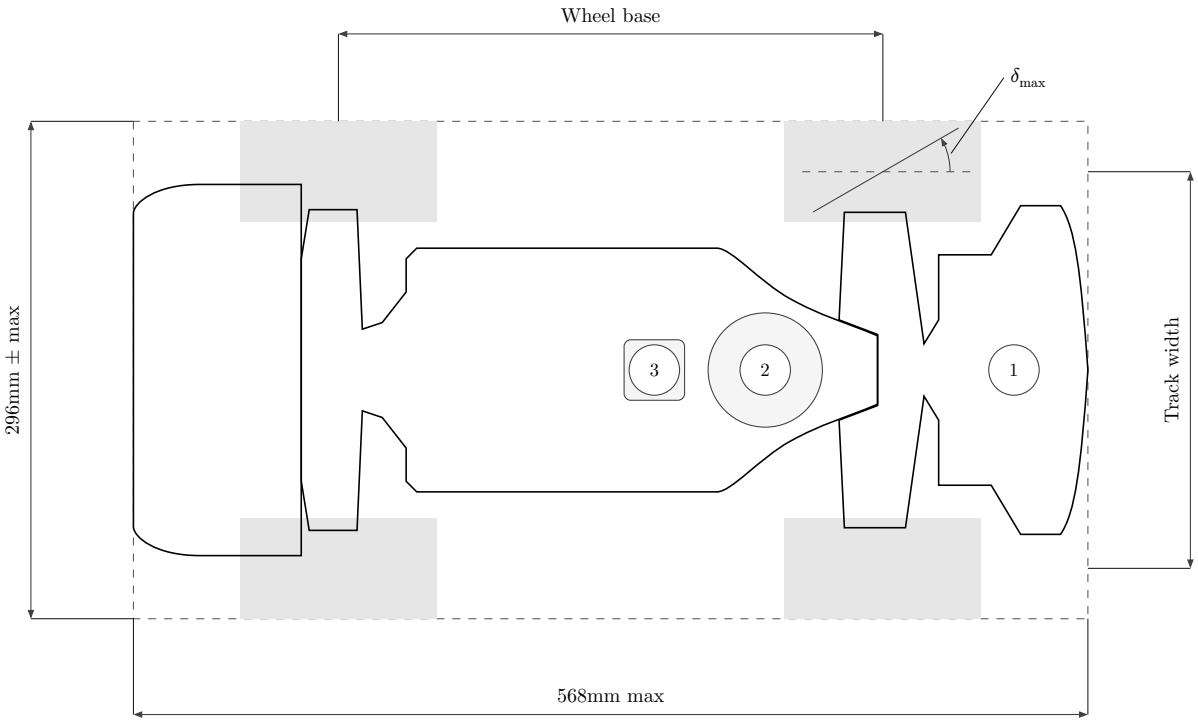


Figure 3.2: Sketch of the RoboRacer vehicle: a 1:10 scale car on a Traxxas chassis

Chapter 4

Simulation environments

4.1. F1Tenth Gym

The F1Tenth Gym [31] is a python package that provides an environment for training and evaluating autonomous racing agents in simulation. It is built on top of the OpenAI Gym framework and features a simple bicycle model, which lumps tyre forces per axle and has no wheel-level states, so quantities such as per-wheel slip are unavailable. The F1Tenth Gym is therefore unsuitable for methods like ours that require per-wheel information.

4.2. AutoDrive

AutoDrive is another simulator for autonomous driving reseach based on Unity and the MLAgents toolkit.

4.3. Unity

Unity is a game engine developed by Unity Technologies which provides a platform wherein simulations can be built and run, simplifying interations with Nvidia's PhysX physics engine and other low-level libraries.

Unity was chosen over the F1Tenth Gym because its `WheelCollider` component exposes the per-wheel slip quantities that are used as privileged information for the auxiliary heads in [WHERE], and because it allows the tyre and suspension parameters to be varied between episodes. It features built-in LiDAR simulation, occupancy grid generation, and an API built for agent training (MLAgents).

4.3.1. Vehicle model

4.3.1.1. The WheelCollider

Unity simulates vehicle dynamics with PhysX. Each wheel of the RoboRacer is modelled as a `WheelCollider`, which simulates the wheel's suspension, rotation and tyre friction [32]. The parameters used are listed in Appendix A.

The tyre model is built on the idea of *slip*: how much the tyre slides over the road instead of rolling on it. A wheel slips when it spins faster than the car moves (wheelspin), turns slower than the car moves (braking), or is dragged sideways (cornering). Slip is expressed as a ratio, so a freely rolling wheel has a slip of 0 and a locked wheel has a slip of 1.

A tyre needs some slip to produce force. The `WheelCollider` computes the force from the slip using a friction curve (Figure 4.1). Force rises with slip until it reaches a peak, the *extremum*, which is the tyre's maximum grip. Beyond the peak, the tyre begins to slide and the force falls to a lower, constant value, the *asymptote*. Pushing the car harder therefore gives less force, not more, which is a regime a racing agent must learn to avoid. The curve is defined separately for the forward direction (accelerating and braking) and the sideways direction (cornering). Note that in real life, at low slip values, the rubber stretches instead of actually sliding on the road. This is modelled by the rising part of the curve at low slip values.

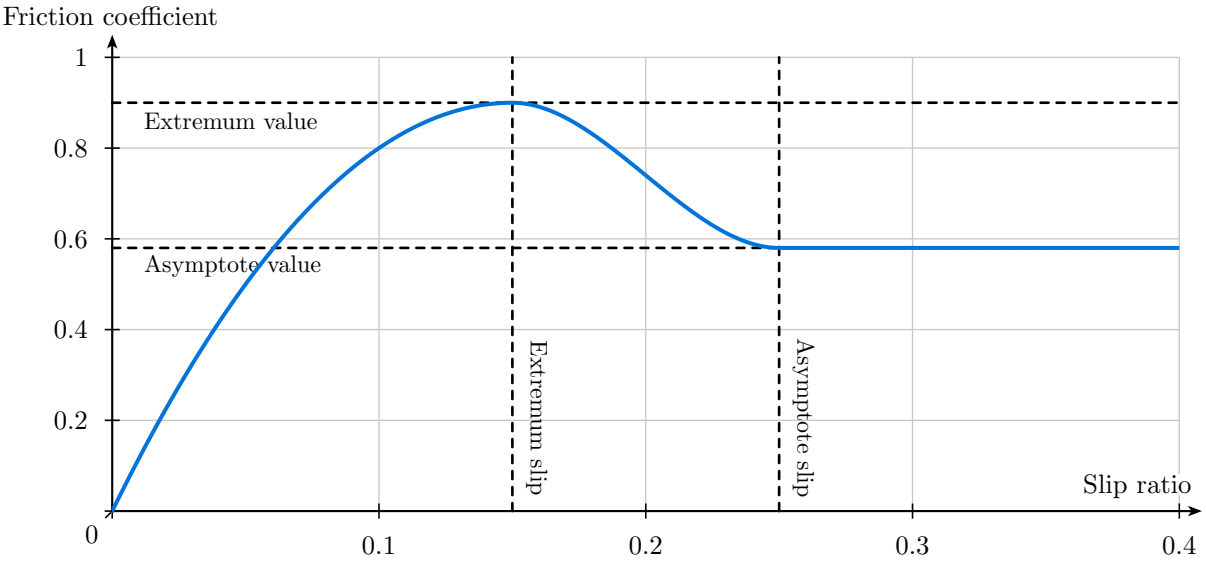


Figure 4.1: Approximated WheelCollider friction curves for the RoboRacer [32][4]

[4]The parameter values detailed in this chart are the default values configured in the AutoDrive simulator for the F1TENTH vehicle's `WheelColliders`

4.3.1.2. Car RigidBody

The car is modelled using a `RigidBody` and various `BoxColliders` to represent the chassis, bumpers, and other parts of the vehicle. The `RigidBody` is responsible for simulating the car's own physics, such as its inertia, and keeping the wheels attached to the car. The `BoxColliders` are used to detect collisions with the track walls and other objects in the environment. The `RigidBody` is configured with a mass of 3.47 kg as in the original AutoDrive simulator.

4.4. MLAgents

4.4.1. Training throughput

Training was performed on a single machine with an AMD Ryzen Threadripper PRO 3995WX CPU (64 cores, 128 hardware threads) and 128 GB of RAM. To make experiments finish in a practical time, we first measured how best to use this hardware, aiming to maximise *aggregate* throughput: environment timesteps collected and learned from per second, summed across all models training simultaneously.

We ran a grid search over the parameters in Table 4.1: concurrent trainers (2, 4, 8, 12), Python threads per trainer (8, 16) and parallel environments per trainer (4, 8, 10, 16). For each configuration, `concurrent_trainers` identical processes were launched at once, each running five PPO iterations (a 256-frame rollout plus four epochs of minibatch updates) with checkpointing and TensorBoard logging disabled. Configurations with more than 8 trainers and 16 environments (>128 concurrent Unity instances) exhausted memory and were excluded, leaving 28 single-run configurations. Throughput per trainer was computed from the mean wall-clock time per iteration:

$$\text{throughput} = \frac{\text{rollout length} \times \text{parallel envs} \times \text{areas per env}}{\text{seconds per iteration}}$$

We report the throughput per trainer, showing how much each model slows as the machine gets busier, and summed across trainers, showing where adding trainers stops paying off.

As shown in Appendix C, summed throughput peaks at 8 trainers (4172 timesteps/s with 8 environments and 16 threads) and falls at 12 (at most 3853 timesteps/s), while per-trainer throughput roughly halves from 2 to 8 trainers. The best settings use $8 \times 16 = 128$ threads in total, matching the hardware thread count; at 12 trainers the 16-thread configurations are oversubscribed and fall behind the 8-thread ones. We therefore chose $(\text{concurrent_trainers}, \text{parallel_envs}, \text{python_threads}) = (8, 8, 16)$, though

its margin over the next-best configurations (4020 and 3962 timesteps/s) is within likely run-to-run variation. Using 8 or 12 areas per environment at this configuration did not help (3844 and 3999 timesteps/s).

By default, Unity spawns `n_cpus` garbage-collector threads; we set this to 8 with the `-gc-helper-count` launch argument for all runs.

Table 4.1: Parameters varied in the training performance grid search.

Parameter	Values considered	Description
`concurrent_trainers`	$\{2, 4, 8, 12\}$	Number of independent models training simultaneously, each in its own process.
`python_threads`	$\{8, 16\}$	Threads available to each trainer's PyTorch, OpenMP, and OpenBLAS thread pools.
`parallel_envs`	$\{4, 8, 10, 16\}$	Number of Unity environment instances per trainer. Each contains 6 training areas, so each instance steps 6 agents in parallel.

4.5. Track generation

During model training, having a variety of tracks is important to prevent overfitting to a single track. This improves generalisation to unseen tracks, allowing the model to learn a more robust policy.

For this purpose, a track generator was implemented in C# in Unity to procedurally generate tracks. The track generator takes the following parameters as input: R - the track radius, n - the number of points to generate, $\theta_{\max}$, the maximum turn angle in a single step, ρ - the mean-reversion factor, s - the mean-reversion exponent, $\varphi_{\max}$ - the maximum heading offset, and q - the probability of flipping the driving direction. The track generator outputs a list of points that define the track.

Algorithm 4.1: Random Closed Track

procedure GENERATE-WALK(n, R, $\theta_{\max}$, ρ, s, $\varphi_{\max}$, q)
  ▷ Initialise the walk on the guide circle
  $L \leftarrow 2\pi\frac{R}{n}$
  $P_0 \leftarrow (R, 0)$
  $h \leftarrow 0$

  ▷ Random walk with a mean-reverting heading offset
  for $i = 1$ to n **do**
    $r \leftarrow \rho + (1 - \rho)\left(\frac{i}{n}\right)^s$
    $u \leftarrow \text{Uniform}(-\theta_{\max}, \theta_{\max})$
    $h \leftarrow \text{clamp}((h + u)(1 - r), -\varphi_{\max}, \varphi_{\max})$

```
    α ← 2πi/n + π/2 + h
    P_i ← P_{i−1} + L(cos α, sin α)
  end
  ▷ Spread the closing error evenly over the lap
  g ← P_n − P_0
  for i = 0 to n − 1 do
    Q_i ← P_i − (i/n)g
  end
  ▷ Randomly flip the driving direction, keeping Q_0 fixed
  if Uniform(0, 1) < q then
    (Q_1, ..., Q_{n−1}) ← (Q_{n−1}, ..., Q_1)
  end
  return Q
end
```

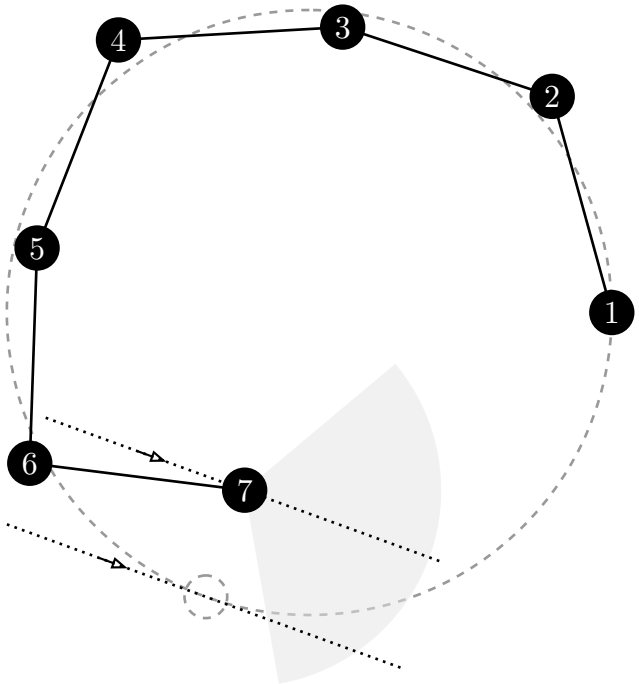


Figure 4.2: Track generation

Then, a Catmull-Rom spline is fitted to the points to create a smooth track centerline. In order to generate the inner and outer boundaries of the track, points are sampled along the centerline to generate a polygonal approximation of the track. The inner and outer boundaries are then generated by insetting/outsetting the centerline polygon by a fixed distance using the Clipper2 library [33], which extends the Vatti polygon clipping algorithm [34].

Once we have the inner and outer boundaries of the track, generating a mesh by extruding a circle or square along the splines is straightforward in Unity. An example track is shown in [ADD PLACE].

Chapter 5

Auxiliary heads

Several auxiliary heads were implemented to predict the following quantities:

Table 5.1: Auxiliary heads implemented to predict the car's forward velocity v_f, lateral velocity v_l, and angular velocity ω.

Auxiliary head(s)	Unit	Description
v_f	m/s	The car's forward velocity
v_l	m/s	The car's lateral velocity (positive is to the left)
ω	rad/s	The car's angular velocity around the verical axis (postive is counter-clockwise)
$v_{s,x}$	m/s	The car's wheel slip velocity per wheel $x \in \{\text{LF}, \text{RF}, \text{LB}, \text{RB}\}$
κ_d	$\frac{1}{m}$	Reciprocal[5] of the radius of curvature of the track at various points d along the track ahead of the car, $d \in \{0.5, 1.0, ..., 8.0\}m$

5.1. The model

The model is a feedforward neural network. Its input is a stack of the 50 most recent LIDAR scans (101 rays each), plus the previous action (steering and throttle). Since the network has no recurrent state, the stacked scans are the only source of temporal information.

The scans are processed by four 1D convolutional layers with ReLU activations. The convolutions slide along the ray axis and treat the 50 time steps as input channels, so the first layer combines the whole history at each angular position and turns it into a set of learned feature channels. The remaining three layers work on these features and only add spatial context along the ray axis. The output is flattened, concatenated with

[5]The reciprocal of the radius of curvature is used to prevent undefined values when the track is straight.

the previous action, and passed through three fully connected layers (128, 72 and 64 units) with ReLU activations and a final layer normalization.

The 64-dimensional output is shared by all heads. The last layer ("out" in Figure 5.1) is a linear layer that produces the outputs of every head at once: 4 for the policy (mean and scale for steering and throttle), 1 for the value estimate, and 27 for the auxiliary prediction targets, which are only used as training signals.

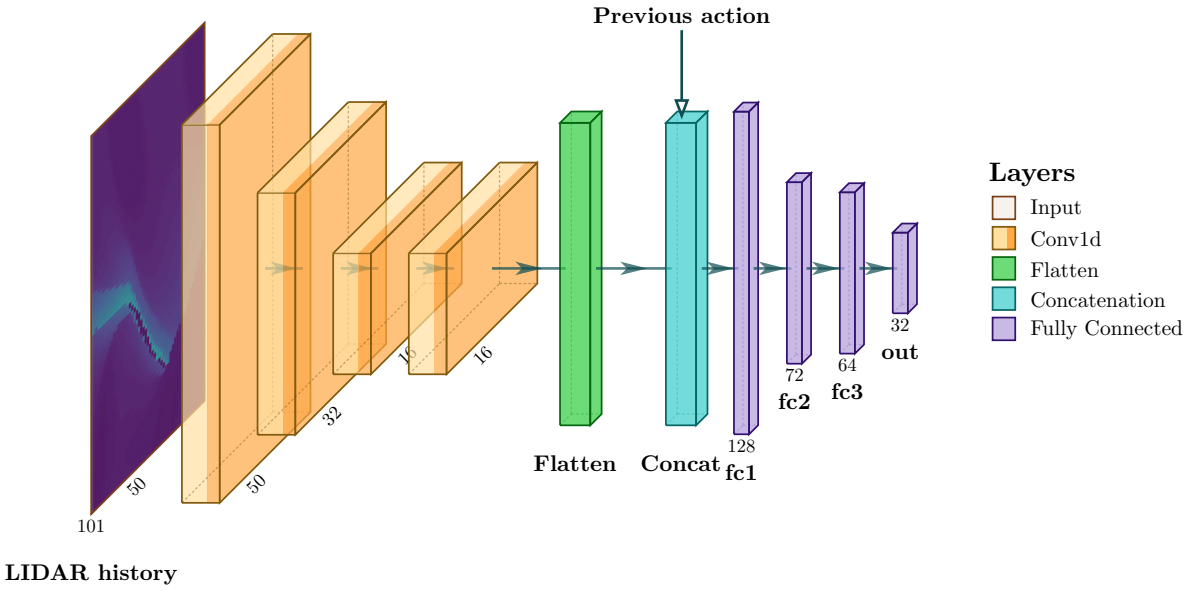


Figure 5.1: Neural network architecture

Chapter 6

Experiments & results

6.1. Baseline performance

6.1.1. Features already encoded in the model

Despite not being trained with the explicit task of auxiliary prediction, we are able to extract a proxy to examine the feature representation in the model.

6.2. Normalisation of ego-agent privledged information

Prior to applying auxiliary learning, the privileged information must be normalised. The auxiliary loss is a sum over the individual prediction targets, so a target with a larger numerical range contributes larger errors and gradients, and can dominate the shared representation at the expense of the others. Training with unnormalised targets makes hyperparameter tuning [35], and training more difficult [36].

Kendall et al. proposes a solution to this problem by learning a log-variance $\log \sigma_i$ for each loss L_i [26]. Because it is expensive to perform hyperparameter searches across the loss weights $\{w_1, ..., w_N\}$,.

Epistemic uncertainty is the uncertainty that comes as a result of lack of training data.

Aleatoric uncertainty is that which cannot be reduced

An alternative method is [37], which dynamically scales the weights of each loss term to balance the $\|\delta_W L_i\|_2$

6.3. Features learnt by the model

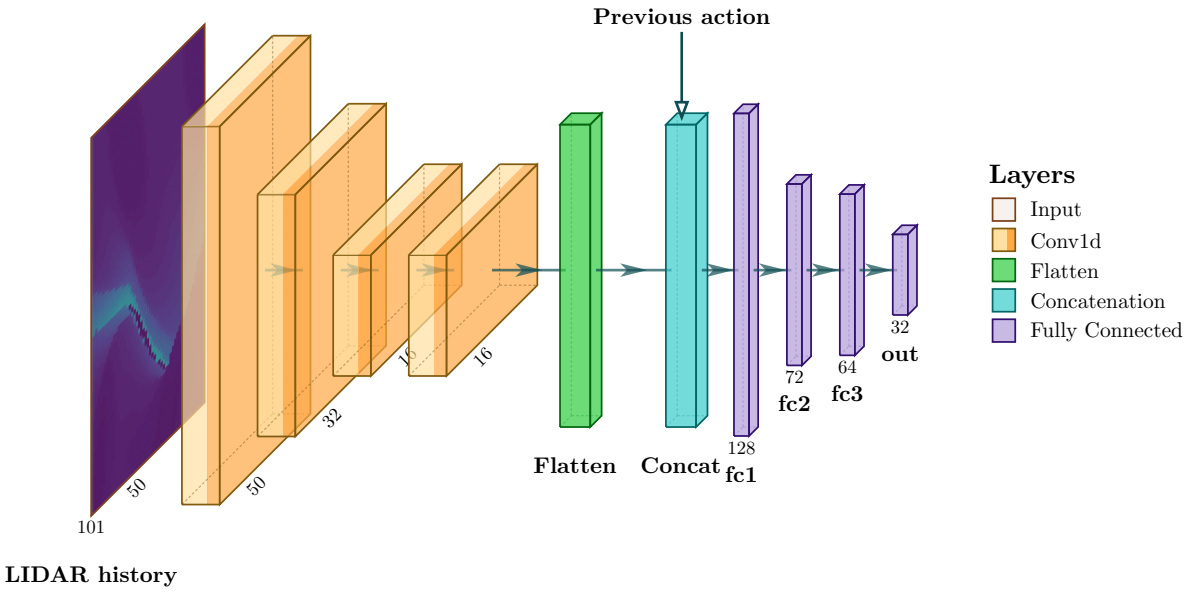


Figure 6.1: Neural network architecture

Setting up a linear regression model to predict the auxiliary targets from the hidden layer (fc3) output of the tunk model, we are able to understand the representation encoded by the model [38]. The results over the training period of three different training configurations at different auxiliary head loss scale weights are shown in Figure 6.2. The three configurations are:

1. trained with all auxiliary heads simultaneously,
2. trained with only the curvature_close heads, and
3. trained without auxiliary heads.

Interestingly, the linear probe model achieves a high R^2 score for the velocity_forward, velocity_lateral, and angular_velocity_y before even training (i.e. just LeCun initialised weights).

[I'm not sure why this happens. I hypothesise that because the velocity is just a linear function of the difference between the length of a single LiDAR ray between two consecutive time steps (i.e. $x_t^{\text{center}} - x_{t-1}^{\text{center}}$), that each layer's projection into a subspace is likely to preserve this linear relationship, and therefore the hidden layer output encodes this information automatically.]

I've only included the curvature_close auxiliary head results in the figure to keep it legible.

More complex features, such as the track curvature at various distances ahead of the car, are not encoded well in the model before training, and the linear proble model is

unable to predict them $R^2_{\text{baseline}} \in [0, 0.2]$ late in training for the baseline model. However, with an auxiliary head, the model is able to learn a representation that encodes this information with $R^2_{\text{curvature_head}} \in [0.4, 0.8]$ later in training for lengths ahead < 3m.

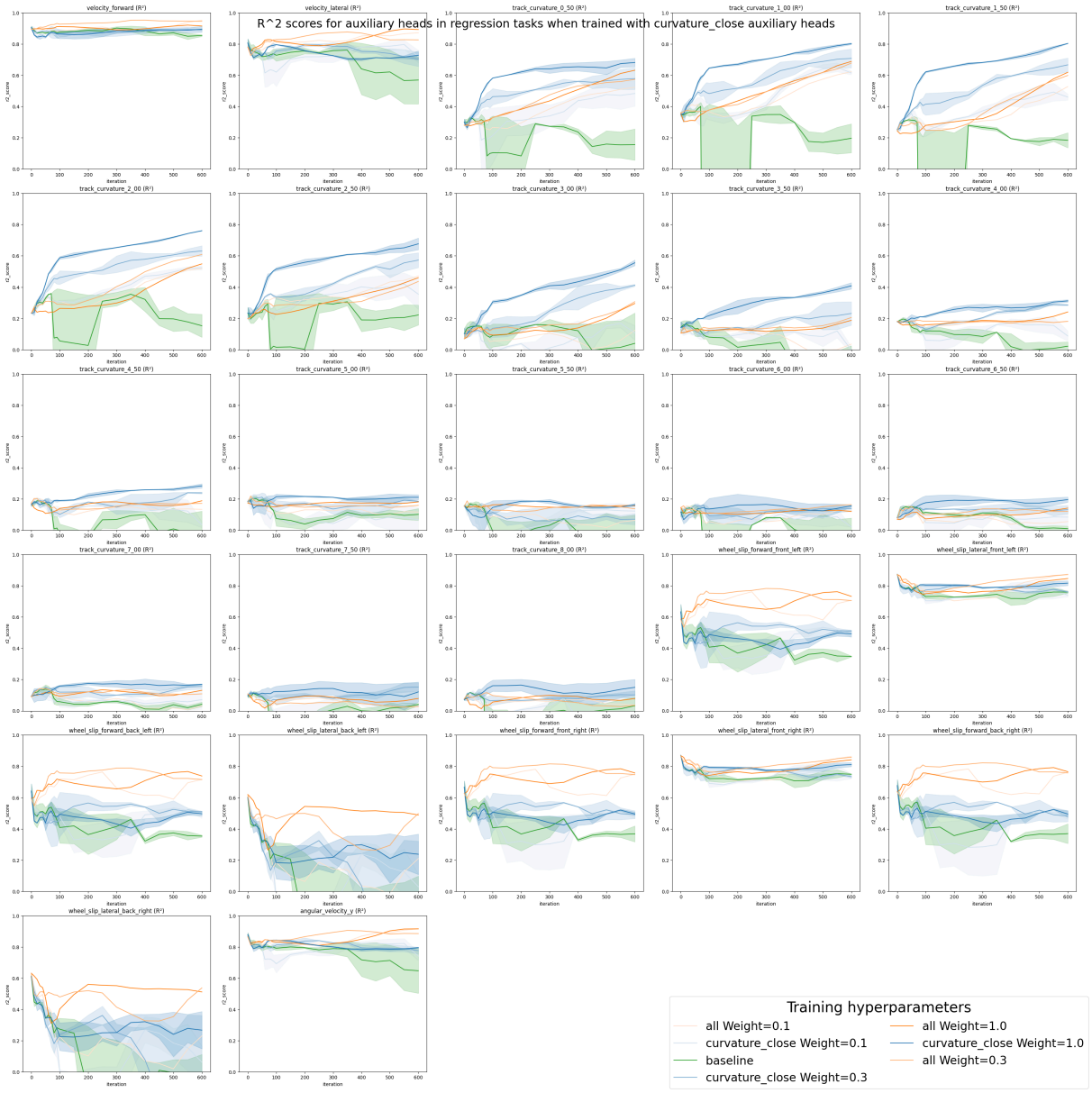


Figure 6.2: R^2 scores for auxiliary head predictions as a function of time. Each subplot is a different auxiliary head. Orange = trained with all aux heads simulatenously, blue = trained with only the curvature_close heads. green = trained without auxiliary heads. Transparency = auxiliary head loss scale weight $w^{(i)}$.

Note that the

Chapter 7

What I did

- **Car always going full throttle + full steering**: This is because tanhnormal saturates at 1, so the model cannot learn when the output is close to 1 or -1. This is a problem because the car is always going full throttle, and the steering angle is often close to 1 or -1. The model cannot learn to output values further away. So, the scale of weights & biases was changed to start in the center
- **LIDAR angle pointing up, causing large lidar values input** - increase wall height, mask "miss" rays to 0
- **Huber loss is invariant** to scale if the parameter beta is updated:

$$L_\beta\left(\frac{x}{\alpha}, \frac{\hat{x}}{\alpha}\right) = \frac{L_{\alpha*\beta}(x, \hat{x})}{\alpha^2}$$

- **Regression based on the hidden layer output of the PPO model**: The PPO model is trained to predict the value function and policy, so the hidden layer output contains information about the state of the environment. This information can be used to predict the auxiliary targets. The regression results show that the hidden layer output contains information about the auxiliary targets, and that this information can be extracted by a simple linear regression model. [38]
- By default, BLAS spawns 128 threads. Did hyperparameter tuning to find the best combination of (num_pytorch_threads, num_parallel_trainers, num_envs_per_trainer). Also Unity spawns 128 GC threads, which can be controlled via `"-job-worker-count", str(num_unity_threads), "-gc-helper-count", str(num_unity_threads)`

Chapter 8

Conclusions and recommendations

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Appendix

Appendix A WheelCollider properties

Table 8.1: Relevant parameters of the WheelCollider model. * = forward & sideways

Property	Description	Value for RoboRacer
`radius`	Radius of the wheel	5.8 cm
`width`	Radius of the wheel	5.8 cm
`mass`	Mass of each wheel; determines its rotational inertia	109 g
`suspension_distance`	Maximum travel of the suspension along the wheel's local Y axis	5 cm
`suspension_spring`	Spring stiffness pulling the wheel toward its target position	500 N/m
`suspension_damper`	Damping that resists the speed of suspension movement	100 (N · s/m)
`extremum_slip`*	Slip at which the friction curve peaks (maximum grip)	0.15 fwd / 0.01 side
`extremum_value`*	Friction at the peak of the curve	0.9 fwd / 1 side
`asymptote_slip`*	Slip beyond which the tyre is fully sliding	0.25 fwd / 0.1 side
`asymptote_value`*	Friction while sliding	0.58 fwd / 0.5 side
`stiffness`*	Multiplier applied to the whole friction curve	0.8 fwd / 1 side

Appendix B Auxiliary Head Regression Results

Appendix C Training Performance Benchmark Results

Table 8.2: Training performance results, sorted by throughput.

Concurrent trainers	Python threads	Parallel envs[6]	Throughput (timesteps/sec)	Speedup vs realtime[7]
8	16	8	4172.37	166.89x
8	16	4	4020.17	160.81x
8	16	8	3999.46	159.98x
8	16	10	3962.12	158.48x
12	8	4	3853.04	154.12x
8	16	8	3843.59	153.74x
12	16	10	3805.19	152.21x
8	8	4	3802.34	152.09x
12	8	8	3751.41	150.06x
12	8	10	3717.67	148.71x
12	16	8	3582.39	143.3x
4	16	10	3428.19	137.13x
12	16	4	3302.99	132.12x
8	8	8	3207.98	128.32x
4	16	8	3201.63	128.07x
8	8	10	3164.72	126.59x
4	16	4	2684.4	107.38x
4	8	4	2414.64	96.59x
2	16	10	2206.84	88.27x
4	8	16	2086.99	83.48x
4	16	16	2075.62	83.02x
2	16	8	2034.77	81.39x
4	8	10	1884.89	75.4x
4	8	8	1830.29	73.21x
2	16	4	1557.77	62.31x
2	8	4	1385.09	55.4x
2	8	16	1122.47	44.9x
2	16	16	1122.1	44.88x
2	8	10	1013.7	40.55x
2	8	8	977.93	39.12x

[6]It was not possible to run 16 parallel envs on 12 trainers, because the machine ran out of memory.
[7]Assuming 25 Hz decision frequency